

D612

Business Process Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A regional insurer with 400 claims staff wants to cut claims turnaround time. The COO has already bought a workflow software license from a leading vendor, and the project kickoff is set for Monday. What should the process team do first?

- A. Remove every manual step from the process.
- B. Configure the purchased license so staff can start using it right away.
- C. Map and measure the current claims process as it actually runs today.
- D. Design the ideal future claims process inside the new workflow tool.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A hospital billing department takes 45 days to collect payment, while similar hospitals take about 20. Leadership wants a clean-sheet rethink of how billing is done, even if job roles and systems change. The department recently moved to a newly renovated floor. Which approach fits?

- A. Automating the existing billing steps as they are.
- B. Redesigning the office layout on the new floor.
- C. Business process reengineering.
- D. Kaizen-style continuous improvement.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A plant manager wants line workers to suggest and test small improvements every week, with no change to job roles or systems. Which approach fits?

- A. Business process reengineering.
- B. Continuous improvement (kaizen).
- C. Outsourcing the production line.
- D. A single redesign followed by no further changes ever.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A process analyst at a mid-sized distributor needs one diagram that shows which department performs each step of order fulfillment and where work passes from one department to the next. The company runs a cloud ERP and ships about 2,000 orders a day. Which diagram fits best?

- A. A SIPOC diagram.
- B. A swimlane (cross-functional) flowchart.
- C. A fishbone (cause-and-effect) diagram.
- D. An organization chart.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A lender that recently won a regional customer-service award maps its 11-step mortgage approval process. From the customer's point of view, which step is value-added?

- A. Sending the customer a survey about the award-winning service.
- B. Re-entering the application data into a second system.
- C. Inspecting the file for errors made at an earlier step.
- D. Underwriting the loan to make the approval decision.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A claims process has three steps in sequence. Intake can handle 60 claims per hour, review 25 per hour, and payment 40 per hour. The intake team just received new high-speed scanners. What is the maximum number of claims per hour the whole process can complete?

- A. 125 claims per hour.
- B. 60 claims per hour.
- C. About 42 claims per hour.
- D. 25 claims per hour.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In the same process (intake 60, review 25, payment 40 claims per hour), management adds a second review station so review can now handle 50 per hour. What is the new maximum throughput?

- A.** 50 claims per hour.
- B.** 25 claims per hour, because the original bottleneck always sets throughput.
- C.** 60 claims per hour.
- D.** 40 claims per hour.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An order-to-ship process takes 10 days from order to shipment. Work that actually changes the order in a way the customer cares about totals 8 hours. The warehouse runs two shifts. Counting each day as 24 hours, what is the process cycle efficiency?

- A.** About 3.3%
- B.** 80%
- C.** 5.0%
- D.** About 96.7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A process has a process cycle efficiency of 5% and a total lead time of 40 hours. How many hours are value-added?

- A.** 8 hours
- B.** 38 hours
- C.** 2 hours
- D.** 35 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A county permit office plans to buy robotic process automation for its 14-step permit process, in which each application is printed, signed, scanned, and re-keyed three times. The IT director has a \$200,000 budget approved. What is the best advice?

- A.** Simplify the process first by removing the duplicate print-scan-rekey loops, then automate what remains.
- B.** Never automate government processes.
- C.** Spend the full \$200,000 now so the budget is not lost.
- D.** Automate all 14 steps as they are to gain speed, then look for improvements later.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A Six Sigma team at a call center has defined the problem and collected baseline data: average handle time is 9 minutes against a 6-minute target. The center serves customers in three time zones. What should the team do next?

- A.** Put a new call script in place right away.
- B.** Analyze the data to find the root causes of long handle times.
- C.** Hire agents in each of the three time zones.
- D.** Write a control plan to hold the gains.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A process team is putting dashboards, standard work documents, and a response plan in place so a newly improved process does not slide back. Which DMAIC stage is this?

- A.** Control
- B.** Define
- C.** Improve
- D.** Measure

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A regional insurer ~~with 400 claims staff~~ wants to cut claims turnaround time. ~~The COO has already bought a workflow software license from a leading vendor, and the project kickoff is set for Monday.~~ What should the process team do first?

Target word: first step in process engineering **Decides:** No one has yet documented how claims actually flow today.

A	Absolute language	Some manual steps add value; you cannot know which until the process is mapped.
B	Camouflage	The license purchase does not decide the method; tools come after the process is understood.
C	Correct	The as-is map and baseline show where time is lost and give a yardstick for any redesign.
D	Inverted operation	Puts the to-be before the as-is; the design would rest on guesses.

2. Answer C Concept 2

A hospital billing department takes 45 days to collect payment, while similar hospitals take about 20. Leadership wants a clean-sheet rethink of how billing is done, even if job roles and systems change. ~~The department recently moved to a newly renovated floor.~~ Which approach fits?

Target word: type of improvement approach **Decides:** “Clean-sheet rethink... even if roles and systems change” signals radical redesign.

A	Dropped element	Leaves the current design in place, dropping the rethink leadership asked for.
B	Camouflage	The new floor does not decide how billing work flows.
C	Correct	BPR is a fundamental, clean-sheet redesign aimed at dramatic gains.
D	Adjacent category	Kaizen makes small, steady changes within the current design, not a clean-sheet rethink.

3. Answer B Concept 2 · transfer

A plant manager wants line workers to suggest and test small improvements every week, with no change to job roles or systems. Which approach fits?

Target word: approach — reversed from radical to incremental **Decides:** Small, ongoing, worker-driven changes.

A	Inverted operation	BPR is radical, one-time redesign, the opposite of what is asked.
B	Correct	Kaizen is small, frequent, incremental change driven by the people doing the work.
C	Adjacent category	A sourcing decision, not a process-improvement method.
D	Absolute language	Contradicts the weekly, ongoing nature of the request.

4. Answer B Concept 3

A process analyst at a mid-sized distributor needs one diagram that shows which department performs each step of order fulfillment and where work passes from one department to the next. ~~The company runs a cloud ERP and ships about 2,000 orders a day.~~ Which diagram fits best?

Target word: diagram type **Decides:** Needs step-level flow plus who does each step and the handoffs.

A	Adjacent category	SIPOC scopes suppliers, inputs, process, outputs, customers at a high level; it does not show step-by-step handoffs.
B	Correct	Lanes show who owns each step; arrows crossing lanes show handoffs.
C	Confusion set	Used to brainstorm root causes, not to map flow.
D	Dropped element	Shows who reports to whom but drops the steps and flow entirely.

5. Answer D Concept 4

A lender ~~that recently won a regional customer service award~~ maps its 11-step mortgage approval process. From the customer's point of view, which step is value-added?

Target word: value-added step **Decides:** Only the underwriting decision changes what the customer is buying.

A	Camouflage	Echoes the award detail; a survey does not move the loan forward.
B	Confusion set	Duplicate data entry is waste; the customer gains nothing.
C	Adjacent category	Inspection and rework feel necessary but only catch errors; they add no value the customer would pay for.
D	Correct	The customer wants a loan decision; underwriting produces it.

6. Answer D Concept 5

A claims process has three steps in sequence. Intake can handle 60 claims per hour, review 25 per hour, and payment 40 per hour. ~~The intake team just received new high-speed scanners.~~ What is the maximum number of claims per hour the whole process can complete?

Target word: process throughput **Decides:** Review, at 25 per hour, is the slowest step.

A	Inverted operation	Adds capacities as if the steps ran in parallel.
B	Camouflage	Intake speed (and its new scanners) does not raise output past the slower review step.
C	Confusion set	$(60 + 25 + 40) / 3$ averages the steps; flow is limited by the slowest one.
D	Correct	Output is capped by the bottleneck step.

7. Answer D Concept 5 • transfer

In the same process (intake 60, review 25, payment 40 claims per hour), management adds a second review station so review can now handle 50 per hour. What is the new maximum throughput?

Target word: throughput after moving the bottleneck **Decides:** Payment, at 40 per hour, becomes the new slowest step.

A	Dropped element	Uses the new review rate but drops the payment step.
B	Absolute language	Once review is expanded, it is no longer the bottleneck.
C	Wrong stem number	Intake capacity, which was never the constraint.
D	Correct	The bottleneck shifts to payment at 40.

8. Answer A Concept 6

An order-to-ship process takes 10 days from order to shipment. Work that actually changes the order in a way the customer cares about totals 8 hours. ~~The warehouse runs two shifts.~~ Counting each day as 24 hours, what is the process cycle efficiency?

Target word: PCE percentage **Decides:** 10 days $\times$ 24 hours = 240 hours of lead time.

A	Correct	$8 / 240 = 0.033$, or about 3.3%.
B	Wrong stem number	$8 / 10$ mixes hours with days.
C	Camouflage	Uses 16-hour days from the two shifts ($8 / 160$); the stem says to count 24 hours.
D	Inverted operation	That is the non-value-added share ($232 / 240$).

9. Answer C Concept 6 • transfer

A process has a process cycle efficiency of 5% and a total lead time of 40 hours. How many hours are value-added?

Target word: solve for value-added time **Decides:** Value-added = PCE $\times$ lead time.

A	Dropped element	$40 / 5$ drops the percent sign and divides.
B	Inverted operation	That is the non-value-added time.
C	Correct	$0.05 \times 40 = 2$ hours.
D	Confusion set	$40 - 5$ treats a percentage as hours.

10. Answer A Concept 7

A county permit office plans to buy robotic process automation for its 14-step permit process, in which each application is printed, signed, scanned, and re-keyed three times. ~~The IT director has a \$200,000 budget approved.~~ What is the best advice?

Target word: sequence of redesign and automation **Decides:** The process is full of duplicate, non-value-added steps.

A	Correct	Redesign before automation; otherwise the robot repeats the waste faster.
B	Absolute language	Automation can help once the process is sound.
C	Camouflage	The budget size does not decide the right sequence.
D	Inverted operation	Automates first and redesigns later, locking in the waste.

11. Answer B Concept 8

A Six Sigma team at a call center has defined the problem and collected baseline data: average handle time is 9 minutes against a 6-minute target. ~~The center serves customers in three time zones.~~ What should the team do next?

Target word: next DMAIC stage **Decides:** Define and Measure are done; Analyze comes next.

A	Dropped element	Skips Analyze and jumps to Improve.
B	Correct	Analyze follows Measure; solutions should target proven causes.
C	Camouflage	Answers the time-zone detail, which does not decide the next stage.
D	Inverted operation	Control is the last stage, after an improvement exists.

12. Answer A Concept 8 · transfer

A process team is putting dashboards, standard work documents, and a response plan in place so a newly improved process does not slide back. Which DMAIC stage is this?

Target word: stage — reversed from activity to name **Decides:** Sustaining gains after improvement.

A	Correct	Control locks in gains with monitoring and standard work.
B	Inverted operation	Define is the first stage, not the last.
C	Adjacent category	Improve builds and tests the change; holding it is Control.
D	Confusion set	Measure sets the baseline before analysis.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Map the as-is before designing the to-be	Document and measure the current process as it actually runs before designing the future process or buying tools.	Starting with the new system or the ideal design.	A project kickoff where a tool or solution is already on the table.
2	Reengineering vs. continuous improvement	BPR is a radical, clean-sheet redesign of a whole process; continuous improvement (kaizen) is small, ongoing, incremental change.	Treating any improvement effort as reengineering.	“Clean sheet,” “fundamental rethink” vs. “small weekly changes.”
3	Choosing the process visual	Swimlane (cross-functional) maps show who does each step and where handoffs occur; SIPOC is a high-level scope view; fishbone is for root cause.	Using a SIPOC or org chart to show step-by-step handoffs.	“Which department does each step” and “where work passes between.”
4	Value-added vs. non-value-added	A step adds value only if the customer would pay for it, it changes the product or service, and it is done right the first time.	Calling inspection, rework, moving, or waiting value-added because it feels necessary.	A list of steps and “from the customer’s view.”
5	The bottleneck sets throughput	A sequential process can go no faster than its slowest step; improving any other step does not raise output.	Averaging or adding step capacities.	Several step capacities in units per hour.
6	Process cycle efficiency	$PCE = \text{value-added time} / \text{total lead time}$, with both in the same units.	Mixing hours with days, or reporting the non-value-added share.	End-to-end time in days and hands-on time in hours.
7	Redesign before you automate	Remove and simplify wasteful steps first, then automate what remains; automating a broken process just makes waste faster.	Buying automation for the process as-is.	An automation purchase for a process full of re-keying and duplicate steps.
8	DMAIC stage order	Define, Measure, Analyze, Improve, Control — root causes are analyzed before solutions, and Control holds the gain.	Jumping from measurement straight to a solution.	“The team has defined... and measured... what next?”

Course scope (public description): Processes form the core of any organization and involve both manual and automated steps. Business Process Engineering introduces how to identify processes, visualize them, and how to design and implement operational methods that promote organizations’ overall efficiency. The course covers common process engineering frameworks, the stages of process engineering present in common frameworks, and introduces tools used to conduct business process reengineering.